

Subject: / /

Q. IP<sub>1</sub> : 202.66.89.99

Subnet Mask : 255.255.255.240

Subnet ID : 202.66.89.96

Subnet No : 6

→

$$\begin{array}{rcl} 99 & \rightarrow & 01100011 \\ 240 & \rightarrow & 11110000 \\ \hline & & 01100000 \\ \hline & & \swarrow \downarrow \\ & & 2^6 \quad 2^5 \end{array} \rightarrow \begin{array}{r} 64 \\ 32 \\ \hline 96 \end{array}$$

→ Subnet mask 1\_\_\_\_.1\_\_\_\_.1\_\_\_\_.11110000

So, we calculate 4 bit in Subnet ID

$$\begin{array}{rcl} 0110 & & \\ 2^3 & 2^2 & 2^1 & 2^0 \\ \hline & & 6 \end{array}$$

← 01100000

Subject:

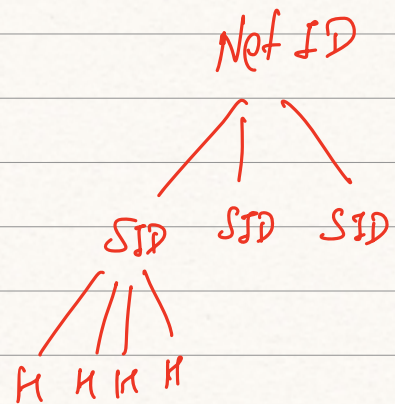
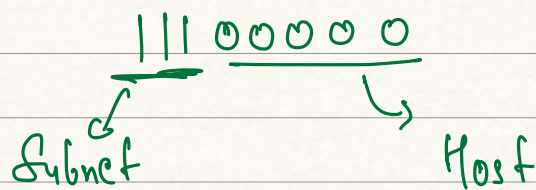
Q. No. of host in each subnet  $2^k - 2$  Host bits

IP : 202.99.89.119

Subnet : 255.255.255.224  
Mask

$$\begin{array}{r} 119 \rightarrow 01110111 \\ 224 \rightarrow \underline{11100000} \\ \hline 01100000 \end{array}$$

Subnet ID :- 202.99.89.96



1st host of that Subnet :-

202.99.89.96      So in Subnet ID

011000000

Subnet (points to 011)      Host (points to 000000)

So, 1st host will be.

011000001



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last host of that subnet

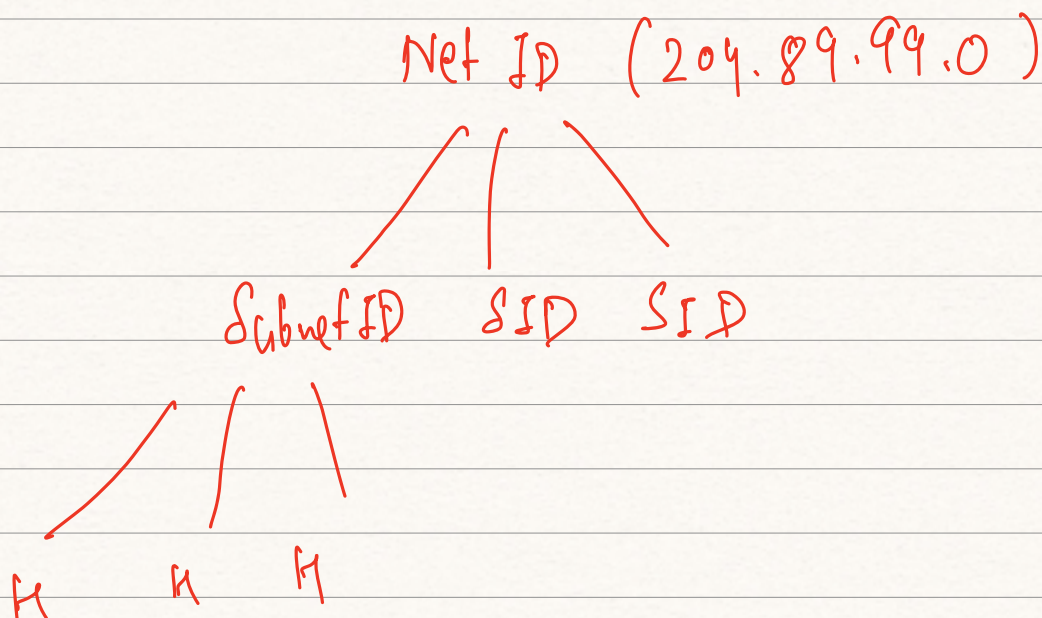
↳ 011 11110

↳ as last one is assigned to DBA/LBA for broadcasting

DBA/LBA of that subnet

↳ 011 11111

Q. IP: 204.89.99.119 <sup>→ class C</sup>  
Subnet: 255.255.255.224  
Mask



Subject: / /

Net Mask: 255.255.255.0

Net ID: 204.89.99.0

Subnet Mask :- 255.255.255.224

$\frac{111}{5} \frac{000000}{11}$

(32)  $\frac{00100000}{\text{first}}$

1st S.I.  $\rightarrow$  32

DBA of network  $\rightarrow$  111 1111

least subnet IP  $\rightarrow$  110 1111

Q. IP, 205.66.77.98

Subnet  
mask 255.255.255.224

224  $\rightarrow$  111 000000

2nd S.I. 010 000000 (64)



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4<sup>th</sup> S.J. 100 00000 (128) //

Q. IP = 202.99.66.89

Subnet Mask = 255.255.255.240

↳ 1111 0000

(i) 2<sup>nd</sup> Host of 4<sup>th</sup> Subnet

4<sup>th</sup> Subnet  $\xrightarrow{2^5}$  0100  $\xrightarrow{2^1}$  0010  
↳ 2<sup>nd</sup> host

202.99.66.66

(ii) 4<sup>th</sup> host of 2<sup>nd</sup> Subnet

$\xrightarrow{2^5}$  0010  $\xrightarrow{2^2}$  0100

202.99.66.36

(iii) 3<sup>rd</sup> host of 1<sup>st</sup> Subnet

$\xrightarrow{2^4}$  0001  $\xrightarrow{2^1, 2^0}$  0011

202.99.66.19

(iv) 1<sup>st</sup> host of 3<sup>rd</sup> Subnet

$\xrightarrow{2^5}$  0011  $\xrightarrow{2^4}$  0001  $\xrightarrow{2^0}$

202.99.66.49

Subject:

Q. Company required 60 Host Mask (Class C) ?  
No. of Host in Subnet  $= 2^6 \rightarrow 64$

Class C mask = 11111111. 11111111. 11111111. 00000000

11000000      N + S + H  
                         24    2    6

Q. Company required 500 Host (Class B)

no. of Host in Subnet      11111111. 11111111. 00000000. 00000000

$2^9 = 512$

↳ 9 bit

N + S + H  
16    7    9

In the above problem if the restriction is Class C ? Can we apply subnetting then we require

Concept of Supernetting

Joining two or more network to form a larger network according to the requirement of user is known as supernetting. During supernetting bits are borrowed from network

In subnet bits are borrowed from host



Subject: / /

Q. One of the address of supernet is

201.55.89.93

Supernet Mask is : 255.255.252.0

Range of Supernet is :-

Supernet IP ——— DBA } Range of  
Supernet IP } Supernet IP

→ 252.0  
Mask  
IP(88)  
1 . 1 . 1111100 . 00000000  
01011001 .  
01011000  
(88)

Supernet IP:-

201.55.88.0

in class C  
Network Supernet Host  
N . N . 1111100 . 00000000

IP 201.55.89.93

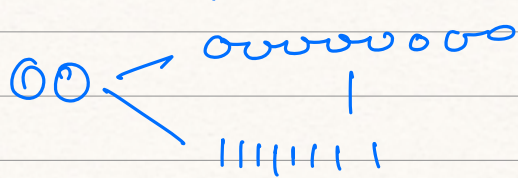
Supernet  
Mask 255.255.252.0

201.55.88.0 → Supernet ID

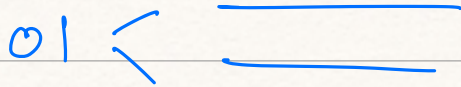
Subject:

Supernet 00

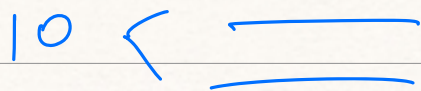
//



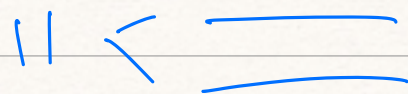
88.0 — 88.255



89.0 — 89.255



90.0 — ~~90~~.255



91.0 — ~~91~~.255

Find the range of all the Supernet ID